

Blockchain Mining Power and Market Quality in Cryptocurrency Trading*

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Abstract

This paper investigates the relationship between aggregate mining power of blockchain and the market quality of cryptocurrency trading platforms. Using historical intraday data of cryptocurrency trading from January to June 2021, we observe that a decrease in the blockchain's aggregate mining power is associated with higher volatility, wider price discrepancies, and higher transaction costs. We also analyze abnormal hashrate and find consistent results. The contribution of this study lies in empirically exploring the connection between blockchain mining and the trading environment of the crypto market, highlighting the importance of blockchain validation process when designing efficient markets for crypto assets.

Keywords: Cryptocurrencies, Blockchain, Hashrate, Market quality

JEL classification: D47, G10, G14

1 Introduction

Cryptocurrencies, a relatively new instrument, have gained significant attention among market participants. Concerns about market quality revolve around the functionality of trading venues, who settle and record transactions on blockchains.¹ As per Nakamoto (2008), the bitcoin blockchain utilizes the Proof-of-Work consensus protocol, where validators engage in a computationally intensive competition to solve numerical puzzles. This validation process rely heavily on the blockchain's aggregate mining power, and alterations in the mining power may impact the overall market quality.

In this paper, we employ panel regressions to explore the relationship between market quality and blockchain aggregate mining power, measured by hashrate. We analyze high-frequency intraday data for Bitcoin, Ether, and Litecoin traded on Coinbase, FTX, and Kraken from January to June 2021. The dependent variable represents market quality measures, while the independent variable is the normalized aggregate hashrate (combining contemporaneous and previous day values). We also consider unexpected changes in mining power by calculating abnormal hashrate, derived from the difference between actual daily hashrate and a smoothed version obtained through the Hamilton (2018) filter. In our regression analysis, we observe a decline in market quality with decreasing blockchain aggregate mining power. More specifically, we document significant increase in volatility, larger price discrepancies, and higher transaction costs when the bitcoin blockchain's aggregate mining power drops. Furthermore, the increase in transaction costs is primary explained by higher adverse selection risk, measured by price impact and Kyle's Lambda, and not by changes in bid-ask spread.

Our paper is the first empirical study exploring the connection between blockchain computing power and market quality in cryptocurrency trading. It combines two strands of literature. The first one focuses on understanding how various characteristics of blockchain, such as mining and network features, influence the pricing of cryptocurrencies. In a theo-

¹For an overview of the crypto-trading infrastructure, see Chen et al. (2023).

retical model, Pagnotta and Buraschi (2018) argue that the blockchain's hashrate and the price of Bitcoin are jointly determined. Pagnotta (2022) further links Bitcoin prices to the security level of the blockchain. Additionally, Easley et al. (2019) present a model that emphasizes the relationship between Bitcoin price, mining rewards, transaction fees, and waiting time. On the empirical front, Liu and Tsyvinski (2021) find that mining characteristics are not priced in cryptocurrency returns. However, Bhambhwani et al. (2021) demonstrate that blockchain hashrate and cryptocurrency prices are cointegrated with mining capacity.² The second strand of literature focuses on examining the market quality provided by cryptocurrency trading platforms. In terms of liquidity, Brauneis et al. (2021) compare the accuracy of different low-frequency liquidity measures for cryptocurrencies. Barbon and Ranaldo (2022) compare transaction costs between centralized platforms and decentralized platforms. Additionally, several studies investigate arbitrage opportunities and price discrepancies among multiple crypto-trading platforms (e.g., Makarov and Schoar (2020); Crépeillère et al. (2023)). Finally Hautsch et al. (2018) present a theoretical study showing that arbitrage boundaries increase with expected latency and latency uncertainty, a finding that our empirical results support.

The subsequent sections of the paper are structured as follows. Section 2 outlines the data and measures used in our analysis. Section 3 presents and discusses the empirical results. Finally, Section 4 serves as the conclusion.

2 Data and variable construction

A key feature of blockchain is its reliance on a consensus protocol, which establishes the way validators reach an agreement on the current state of the ledger. As outlined in Nakamoto (2008), the bitcoin blockchain implements the Proof-of-Work consensus proto-

²Several papers (e.g., Cong et al. (2021) and Sockin and Xiong (2023)) focus on the production side of the cryptocurrency and show that the evolution of cryptocurrency prices is linked to the marginal cost of production.

col, where validators enter into a computationally intensive competition to solve numerical puzzles by continuously guessing the solution of hash functions. This validation process is also generally known as mining and depends on the blockchain's aggregate mining power. To capture the blockchain's mining power, we collect daily hashrate from Coin Metrics. The hashrate serves as an indirect measure of the aggregate computing capacity of the blockchain network. It is estimated based on the number of blocks mined during a given day and the mining difficulty.

We supplement our analysis with data from Kaiko, a crypto data vendor.³ We obtain historical intraday order book and trading data for three major cryptocurrencies—Bitcoin, Ether, and Litecoin—all traded against the U.S. dollar. The data spans from January to June 2021 and is sourced from three major crypto trading platforms: Coinbase, Kraken, and FTX. For the order book data, we get the best bid and ask price at a twice per minute frequency on each platform. Then, for every minute and trading platform, we compute the volume-weighted bid and ask price to form minute-level observations. Moreover, we have the minute-level price and trading information, including last price and trading volume.

Our goal is to examine the correlation between market quality and blockchain mining power. We estimate minute-level measures for market efficiency and liquidity. As the mining hashrate is reported daily, we aggregate the minute-level measures to a daily frequency for each crypto platform. More specifically, we compute average values across the three crypto platforms. The arbitrage indexes, which involve data from multiple platforms, are measured at a daily level, not separated by trading platforms. This methodology ensures a consistent daily-level dataset for our analysis.

[Table 1 about here.]

We calculate a total of eight measures capturing cryptocurrency return characteristics and market quality: return, volatility, trade-based arbitrage index, quote-based arbi-

³Kaiko's data is widely employed in empirical studies in the field. For example, see Biais et al. (2023); Makarov and Schoar (2020); Bhambhwani et al. (2021); Barbon and Rinaldo (2022); Crépeillère et al. (2023).

trage index, quoted spread, effective spread, price impact, and Kyle (1985)'s Lambda. Appendix A details their construction. Table 1 provides summary statistics for the blockchain hashrate and the market quality measures. Comparing the sample means across the three cryptocurrencies, we note that Bitcoin generally displays more favorable market trading conditions compared to Ether and Litecoin.

3 Hashrate and market quality

To analyze the relationship between blockchain mining capacity and the market quality provided by the crypto trading platforms, we run the following cryptocurrency-day level panel regression.

$$MarketQuality_{i,t} = \beta_1 norm(HashRate)_t + \beta_2 norm(HashRate)_{t-1} + \alpha_i + \varepsilon_{i,t} \quad (1)$$

$MarketQuality_{i,t}$ represents the market quality measure of interest for cryptocurrency i on day t . $HashRate_t$ denotes the blockchain mining capacity measured by the hashrate on day t . $norm()$ is the normal standardization function. We include cryptocurrency fixed effects as α_i to account for the differences across the instruments. In order to account for potential correlation within each cryptocurrency, we cluster the standard errors ($\varepsilon_{i,t}$) at the cryptocurrency level.

Table 2 presents the estimation results. Notably, we observe statistically significant and negative coefficients (β_2) for several market quality measures. Specifically, return volatility, the quote-based arbitrage index, price impact, and Kyle's Lambda exhibit significant negative relationships with the previous day's hashrate (p -value < 0.1 for return volatility, the quote-based arbitrage index, and price impact; p -value < 0.05 for Kyle's Lambda). A decrease in the blockchain's aggregate mining power is associated with increased return volatility, price discrepancy, and transactions costs in terms of price impact on the following day. These findings suggest that a decrease in the previous day's hashrate has an adverse

effect on the quality of cryptocurrency trading on the subsequent day.

[Table 2 about here.]

Moreover, we also examine the unexpected change in blockchain hashrate to better understand its impact on the cryptocurrency trading market. Our intention is to alleviate the concern that market participants might predict the changes in mining power, act accordingly, and affect market quality, confounding our results. Following Makarov and Schoar (2020)'s measure of buying pressure, which is the difference between the actual bitcoin price and the smoothed bitcoin price, we construct the abnormal hashrate (*abHashRate*). This measure represents the difference between the realized hashrate and the smoothed hashrate, which is estimated by employing the filter proposed by Hamilton (2018).⁴ More specifically, we estimate an autoregressive model with four lags for the hashrate, using the previous 90 days' hashrate data.⁵ Based on the regression results, we predict the hashrate for the following day (i.e., $t + 1$) as the smoothed hashrate. Finally, the abnormal hashrate is computed as the difference between the actual hashrate and the smoothed hashrate. The estimation is formulated bellow.

$$HashRate_t = c + \sum_{\tau=1}^4 \beta_{\tau} HashRate_{t-\tau} + \varepsilon_t \quad (2)$$

$$abHashRate_{t+1} = HashRate_{t+1} - \widehat{HashRate}_{t+1} \quad (3)$$

We standardize the abnormal hashrate to ensure that the variable is on a comparable scale. We then re-estimate the panel regression models and present the results in Table 3. In Panel A, the estimated coefficients of both the contemporaneous hashrate and the lagged hashrate are negative and statistically significant for return volatility, and the coefficient of lagged hashrate one quote-based arbitrage index is also negative and statistically

⁴Makarov and Schoar (2020) use the Hodrick and Prescott (1997) filter for the unexpected buying pressure, but we use Hamilton (2018) filter to avoid the drawbacks pointed out in this paper.

⁵Results are consistent with a 30-day or 180-day estimation window.

significant. A drop in hashrate is associated with notable increase in volatility and price discrepancy.

In Panel B, the result suggests that a decrease in hashrate is associated with higher transaction cost, measured by effective spread, both contemporaneously and for the subsequent day. Transaction costs consist of profits from liquidity supply, measured by quoted bid-ask spread, and adverse selection risk, measured by price impact and Kyle's Lambda. The primary driver of change in transaction cost is price impact and Kyle's Lambda, as the estimated coefficients are negative and statistically significant. We do not observe significant estimated coefficient for quoted spread.

[Table 3 about here.]

In summary, the results indicate that bitcoin blockchain's mining power have a significant association with the market quality of cryptocurrency trading platforms. Specifically, decrease in mining capacity is linked to higher volatility, increased price discrepancy across platforms, and higher transaction costs. These findings align with the expectation that blockchain mining capacity affects the functionality of transaction validation, which in turn influence the price efficiency and liquidity of cryptocurrency markets. Moreover, our results contribute to the existing literature on blockchain characteristics and their impact on cryptocurrency pricing, supporting the notion that blockchain features play a crucial role in shaping the dynamics of cryptocurrency markets (Easley et al. (2019); Bhambhani et al. (2021); Datta and Hodor (2022); Pagnotta (2022)).

4 Conclusion

With the surge in popularity of cryptocurrencies, concerns regarding the trading quality offered by cryptocurrency trading platforms have surfaced. Regulatory bodies like the U.S. SEC and the U.K. FCA have heightened their scrutiny of cryptocurrency trading, necessi-

tating an investigation into the market quality provided by crypto trading platforms and the factors influencing their proper functioning.

In this study, we explore the relationship between blockchain's aggregate mining power and the market quality offered by crypto trading platforms. Analyzing historical intraday data of three cryptocurrencies traded on three platforms from January to June 2021, we calculate various market quality measures, including price discrepancy and transaction costs. Through panel regression analysis, we observe that a decrease in blockchain's aggregate mining power is linked to higher volatility and wider cross-platform price discrepancies. Furthermore, we find that the transaction costs increase when the blockchain's aggregate mining power declines.

Our study makes a valuable contribution to the academic literature as the first empirical examination of the relationship between blockchain mining and the crypto market trading environment. The findings also shed light on the market design of crypto trading, where trading settlement occurs on the blockchain and relies on the validation process beyond the control of trading providers. Nonetheless, we discover that the aggregate mining power can impact the trading quality offered by the platforms. Therefore, regulators and stakeholders should consider this unique feature of blockchain technology when evaluating the design of a fair and well-functioning market for crypto assets.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used ChatGPT in order to check for grammar errors. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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Appendix A: Market quality measure construction

For each crypto platform j and cryptocurrency i , we utilize minute-level data (denoted by subscript τ) to estimate daily market quality measures (denoted by subscript t). The specific methods used for estimation are described below, and we suppress the subscripts i and j in the formulas. To facilitate easier interpretation and comparison with hash rates, we scale certain variables using a multiplier. Finally, the daily market-wide measures for each cryptocurrency are obtained by calculating the average across the three crypto platforms.

- Return (r)

Return is the natural logarithmic difference between the beginning and closing quote midpoint, which is the average of the best bid and ask prices $price_t^{mid} = (price_t^{ask} + price_t^{bid})/2$

$$r_t = \log(price_t^{mid}) - \log(price_{t-1}^{mid})$$

- Volatility (σ)

Return volatility is measured by the realized volatility, which is the squared root of the sum of the intraday squared returns

$$\sigma_t = \sqrt{\sum_{\tau} r_{\tau}^2}$$

- Quoted spread (QS)

The percentage quoted spread is the difference between the best ask price ($price^{ask}$) and the best bid price ($price^{bid}$) of each order book snapshot, divided by the quote midpoint ($price^{mid}$). The daily quoted spread is the simple average of the intraday

measure. We scale this measure by 1,000

$$QS_{\tau} = \frac{price_{\tau}^{ask} - price_{\tau}^{bid}}{price_{\tau}^{mid}}$$

$$QS_t = \overline{QS_{\tau}} \times 1000$$

- Effective spread (ES)

The effective spread is calculated as follow. The daily effective spread is the simple average of the intraday measure. We scale the measure by 1,000

$$ES_{\tau} = \frac{2 \times |price_{\tau}^{trade} - price_{\tau}^{mid}|}{price_{\tau}^{mid}}$$

$$ES_t = \overline{ES_{\tau}} \times 1000$$

where $price_{\tau}^{trade}$ is the transaction price recorded during the same minute.

- Price impact (PI)

We first determine the trading indicator ($D=+1$ for buyer-initiated trades and $D=-1$ for seller-initiated trades) applying the Lee and Ready (1991) algorithm. Then we compute the percentage change in the quoted mid price after taking into account the trade direction. The daily price impact is the simple average of the intraday measure. We scale the measure by 1,000

$$PI_{\tau} = D_{\tau} \times \frac{Price_{\tau+1}^{mid} - Price_{\tau}^{mid}}{Price_{\tau}^{mid}}$$

$$PI_t = \overline{PI_{\tau}} \times 1000$$

- Kyle's Lambda (λ)

Kyle (1985)'s Lambda measures the cost, in terms of price movement, of taking liquidity and is an inverse measure of liquidity. To compute the daily Kyle's Lambda,

we estimate the OLS coefficient λ using the daily intraday observations. We scale the measure by 10,000

$$r_{\tau} = c + \lambda D_{\tau} \log(\text{volume}_{\tau} \times \text{price}_{\tau}^{\text{trade}}) + \varepsilon_{\tau}$$

$$\lambda_t = \hat{\lambda} \times 10000$$

In addition to these market quality measures, we also calculate two arbitrage indexes to evaluate market efficiency. The trade-base arbitrage index quantifies the potential for arbitrage by comparing the highest traded price with the lowest traded price observed across multiple trading venues during each minute. On the other hand, the quote-base arbitrage index assesses arbitrage opportunities by examining the highest bid price and the lowest ask price offered by the trading venues on a minute-to-minute basis. The construction of these two indexes is outlined below.

- Trade-based arbitrage index (Arb^{trade})

We construct the trading price-based arbitrage index following Makarov and Schoar (2020). For every minute during our sample period, we take the highest price across all platforms and divide it by the minimum price. Then we average the arbitrage index at the daily level.

$$Arb_{\tau}^{\text{trade}} = \max(\text{price}_{\tau}^{\text{trade}}) / \min(\text{price}_{\tau}^{\text{trade}})$$

$$Arb_t^{\text{trade}} = \overline{Arb_{\tau}^{\text{trade}}} \times 100$$

- Quote-based arbitrage index (Arb^{quote})

In addition to the trade-base arbitrage index, we also construct a bid-ask spread-based arbitrage index. In an arbitrage-free market, the ask price should always be greater than the bid price for the same financial instrument across different trading venues. This condition ensures that arbitrageurs cannot exploit price discrepancies

and make riskless profits. To capture potential arbitrage opportunities, we calculate an arbitrage index for each minute by dividing the highest bid price by the lowest ask price, but only when the bid price is greater than the ask price. If the bid price is not greater than the ask price, we assign a value of 1 to indicate the absence of arbitrage opportunity. Finally, we take the daily average of the arbitrage index to summarize the overall level of potential arbitrage in the market.

$$Arb_{\tau}^{quote} = \max(\max(price_{\tau}^{bid})/\min(price_{\tau}^{ask}), 1)$$

$$Arb_t^{quote} = \overline{Arb_{\tau}^{quote}} \times 100$$

Table 1. Summary statistics

This table reports summary statistics of blockchain aggregate hashrate (Panel A) and market quality measures (Panel B). The sample period is January to June 2021. The construction of the measures is described in the Appendix.

Panel A: Hashrate

	N	mean	std
Hashrate	180	1.5081×10^8	2.1741×10^7

Panel B: Market quality

	N	mean	std	N	mean	std
	Return			Volatility		
btc-usd	180	0.0008	0.0493	180	0.0538	0.0255
eth-usd	180	0.0047	0.0681	180	0.0677	0.0385
ltc-usd	180	-0.0003	0.0755	180	0.0794	0.0394
	Arbitrage Index (Trade)			Arbitrage Index (Quote)		
btc-usd	180	100.0582	0.0315	180	100.0708	0.0390
eth-usd	180	100.0740	0.0345	180	100.0768	0.0451
ltc-usd	180	100.1599	0.0811	180	100.0648	0.0625
	Quoted Spread			Effective Spread		
btc-usd	180	0.0811	0.0683	180	1.3995	0.7615
eth-usd	180	0.1748	0.1172	180	1.7556	0.9819
ltc-usd	180	0.7478	0.4349	180	2.0880	1.0287
	Price Impact			Kyle		
btc-usd	180	0.1502	0.1165	180	0.6836	0.3081
eth-usd	180	0.1947	0.1725	180	0.8961	0.4336
ltc-usd	180	0.1324	0.1854	180	1.4184	0.7063

Table 2. Realized hashrate

This table reports the panel regression results. The dependent variables are, in Panel A, cryptocurrency return, return volatility, the trade-base arbitrage index and the quote-base arbitrage index, in Panel B the quoted spread, the effective spread, price impact, and Kyle (1985)'s Lambda. These measures are computed according to the description in the Appendix. The independent variable is the actual blockchain hashrate. We also standardize the hashrate. We run the regressions for both the contemporaneous hashrate and lagged hashrate separately. Our sample covers three cryptocurrencies (bitcoin, ether, and litecoin) traded on three crypto trading platforms (Coinbase, FTX, and Kraken) from January to June 2021. We include cryptocurrency fixed effects. The standard errors are clustered by cryptocurrency and are reported in parenthesis. *Note:* *p<0.1; **p<0.05; ***p<0.01

Panel A

	Return (1)	Volatility (2)	Arbitrage Index (Trade) (3)	Arbitrage Index (Quote) (4)
HashRate _t	0.002 (0.003)	-0.000 (0.001)	0.011 (0.004)	0.010 (0.005)
HashRate _{t-1}	0.007 (0.003)	-0.005* (0.002)	-0.015 (0.005)	-0.016* (0.005)
Crypto FE	Yes	Yes	Yes	Yes
Clustering	Crypto	Crypto	Crypto	Crypto
Observations	540	540	540	540
R ²	0.006	0.089	0.415	0.026

Panel B

	Quoted Spread (1)	Effective Spread (2)	Price Impact (3)	Kyle (4)
HashRate _t	0.026 (0.012)	-0.068 (0.041)	0.013 (0.007)	-0.039 (0.019)
HashRate _{t-1}	-0.007 (0.004)	-0.111 (0.058)	-0.020* (0.005)	-0.065** (0.014)
Crypto FE	Yes	Yes	Yes	Yes
Clustering	Crypto	Crypto	Crypto	Crypto
Observations	540	540	540	540
R ²	0.559	0.093	0.029	0.277

Table 3. Abnormal hashrate

This table reports the panel regression results. The dependent variables are, in Panel A, cryptocurrency return, return volatility, the trade-base arbitrage index and the quote-base arbitrage index, in Panel B the quoted spread, the effective spread, price impact, and Kyle (1985)'s Lambda. These measures are computed according to the description in the Appendix. The independent variable is the abnormal blockchain hashrate, which is the difference between the actual hashrate and the smoothed hashrate based on Hamilton (2018). We also standardize the abnormal hashrate. We run the regressions for both the contemporaneous abnormal hashrate and lagged abnormal hashrate separately. Our sample covers three cryptocurrencies (bitcoin, ether, and litecoin) traded on three crypto trading platforms (Coinbase, FTX, and Kraken) from January to June 2021. We include cryptocurrency fixed effects. The standard errors are clustered by cryptocurrency and are reported in parenthesis. *Note:* * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Panel A

	Return (1)	Volatility (2)	Arbitrage Index (Trade) (3)	Arbitrage Index (Quote) (4)
$abHashRate_t$	0.003 (0.001)	-0.001** (0.000)	0.002 (0.001)	0.000 (0.001)
$abHashRate_{t-1}$	0.006** (0.001)	-0.003*** (0.000)	-0.004 (0.002)	-0.007* (0.002)
Crypto FE	Yes	Yes	Yes	Yes
Clustering	Crypto	Crypto	Crypto	Crypto
Observations	540	540	540	540
R^2	0.025	0.095	0.414	0.045

Panel B

	Quoted Spread (1)	Effective Spread (2)	Price Impact (3)	Kyle (4)
$abHashRate_t$	0.017 (0.008)	-0.039* (0.009)	-0.001 (0.002)	-0.015* (0.005)
$abHashRate_{t-1}$	0.012 (0.008)	-0.044** (0.005)	-0.009* (0.003)	-0.019** (0.002)
Crypto FE	Yes	Yes	Yes	Yes
Clustering	Crypto	Crypto	Crypto	Crypto
Observations	540	540	540	540
R^2	0.564	0.092	0.032	0.272